Research Proposal

How information systems are transforming healthcare in the United Arab Emirates with an emphasis on the integration of health information exchanges

Integration focus: Riayati • Malaffi • NABIDH

Narrated presentation transcript (~15 minutes)

Context and background

Today I present a research proposal examining how information systems are transforming healthcare in the United Arab Emirates, with an emphasis on the integration of health information exchanges. The UAE has invested heavily in digital health infrastructure, moving from fragmented, paper-based documentation towards interoperable electronic medical records and cross-organisational platforms. Abu Dhabi's Malaffi, launched in 2019, and Dubai's NABIDH are emblematic initiatives intended to enable meaningful, real-time exchange of patient information, while the federal National Unified Medical Record programme, Riayati, has been publicly reported as providing national-level linkage across emirates. (Department of Health – Abu Dhabi, 2019; NABIDH, 2025; Department of Health – Abu Dhabi, 2023)

Motivation and evidence gap

The motivation for this study is that the scale of these programmes is often presented as inherently beneficial, yet the empirical evidence base in the UAE remains methodologically uneven. For instance, descriptive facility-level analyses demonstrate high adoption and increasing maturity of EMR systems in Dubai, including comprehensive hospital implementation and extensive uptake in the private sector. However, cross-sectional designs cannot isolate whether observed improvements, such as fewer documentation errors or more efficient workflows, are attributable to information systems rather than concurrent organisational change. (Abdulrahman et al., 2024; Menachemi and Collum, 2011)

Socio-technical insight

At the same time, socio-technical research indicates that technology adoption is contingent on staff capability, organisational support and perceived usefulness. In a UAE study integrating UTAUT and TAM with structural equation modelling and neural network techniques, facilitating conditions and behavioural intention were strong predictors of EMR use, suggesting that investments in infrastructure alone are insufficient without training, governance and workflow alignment. This insight is particularly relevant for HIE integration, where cross-organisational coordination and trust in shared data are essential. (Almarzouqi et al., 2022)

Research problem

My research problem therefore concerns the gap between large-scale interoperability ambitions and robust evidence of measurable transformation outcomes. Public reporting has described the integration of Riayati, Malaffi and NABIDH as enabling nationwide exchange, including extensive record volumes and broad access across facilities. Such integration creates a natural ‘policy intervention point’ for evaluating whether interoperability translates into reduced duplicated diagnostics, better continuity in mobile populations, and improvements in safety proxies. Yet, without quasi-experimental evaluation, the magnitude and distribution of benefits remain uncertain, and concerns about privacy and trust may reduce effective use. (Department of Health – Abu Dhabi, 2023; WAM, 2023; Alkaabi and Elsori, 2025)

Policy focus and intended improvements

In this proposal, three interrelated policy initiatives provide the empirical setting: Riayati (the National Unified Medical Record programme), Malaffi (Abu Dhabi's health information exchange), and NABIDH (Dubai's health information exchange). Although each programme has its own participation and governance arrangements, this study is primarily focused on the 2023 national integration policy that links these platforms. This integration policy is the most directly aligned with measurable improvements in cross-emirate care continuity, patient safety, and care delivery efficiency because it enables clinicians to access longitudinal records when patients transition between emirates and providers.

Theory of change for continuity, safety, and efficiency

The integration of health information exchanges is expected to improve continuity of care by reducing information gaps at transitions, enabling faster access to medication histories, allergies, and prior investigations, and supporting more reliable referrals and follow-up. It is expected to improve patient safety by strengthening medication reconciliation, reducing duplicate prescribing, and enabling earlier detection of contraindications through more complete information. It is expected to improve efficiency by reducing duplicated laboratory testing and imaging, shortening administrative workflows, and improving the timeliness of clinical decision-making. However, these benefits are not automatic, because they depend on data quality, clinician use at the point of care, and robust governance that sustains trust.

Overarching research question

The overarching research question is: to what extent has the integration of UAE HIE platforms improved continuity of care, patient safety and resource efficiency, and what socio-technical factors explain variation in realised benefits across organisations? (Department of Health – Abu Dhabi, 2023; Almarzouqi et al., 2022)

Evaluation focus: continuity of care • patient safety • resource efficiency • socio-technical variation

Aim and objectives

The aim is to deliver a mixed- methods evaluation that combines causal inference, socio- technical explanation and a computing artefact that can support governance. The first objective is conceptual: I will develop a UAE- specific model linking HIE integration to information quality, clinician adoption, and clinical and operational outcomes. The second objective is quantitative: I will quantify changes in duplicated diagnostics and selected utilisation and safety proxies before and after integration using quasi- experimental methods. The third objective is interpretive: I will explore professional and patient perspectives to understand mechanisms, barriers and enablers. Finally, the fourth objective is developmental: I will create a prototype dashboard that operationalises interoperability and data- quality indicators. (Hong et al., 2018; Department of Health – Abu Dhabi, 2023; NABIDH, 2025)

Key literature strand 1

In terms of key literature, several strands are directly relevant. First, evidence on EMR adoption in Dubai indicates strong coverage and increasing maturity. Abdulrahman and colleagues assessed 2,089 healthcare facilities and reported that hospitals had implemented EMRs, with extensive adoption among private clinics, using HIMSS EMRAM benchmarking. This demonstrates readiness for interoperability; however, it is not, in itself, evidence that interoperability improves outcomes. (Abdulrahman et al., 2024)

Key literature strands 2–3

Second, technology acceptance lit erature provides a mechanism for heterogeneous effects. Almarzouqi and colleagues show that facilitating conditions and perceived usefulness predict EMR use, implying that even when HIE integration is technically available, the degree of clinician engagement may differ markedly across sites. (Almarzouqi et al., 2022)

Third, evidence on telemedicine and digital inequalities highlights that awareness and usability shape uptake of digital services. In a nationwide survey, 31.3 per cent of respondents used telemedicine during the COVID- 19 period, while non- use was frequently attributed to not knowing that telemedicine existed or not knowing how to use it . Although telemedicine is not the focus of this proposal, these findings reinforce that technology impact s are mediat ed by lit eracy, access and t rust, which are also relevant for HIE-enabled services. (Al Meslamani et al., 2022)

Key literature strand 4

Finally, qualitative research on AI and telemedicine in the UAE reports perceived benefits alongside persistent concerns about data protection, governance and trust. These themes are directly applicable to HIE integration, which relies on cross-organisational sharing and a credible privacy and security framework. (Alkaabi and Elsori, 2025)

Key recurring themes in the UAE evidence base include data protection, governance, and trust.

Barriers to interoperability: competing perspectives

Prior research presents competing perspectives on why interoperability benefits are often uneven. A technical readiness perspective emphasises standards alignment, semantic consistency, and data quality as the primary constraints, arguing that poor data and incompatible systems limit clinical usefulness. An organisational readiness perspective argues that leadership support, training, workflow integration, and incentives determine whether interoperability is used meaningfully, even when it is technically available. A trust and governance perspective emphasises privacy, consent, accountability, and perceived legitimacy, arguing that weak governance can suppress use and undermine data sharing across organisations. This proposal increases critical depth by explicitly comparing these perspectives and testing which barriers best explain variation in realised outcomes after integration.

Operationalising barriers and strengthening synthesis

Analytically, the quantitative strand will estimate average effects and heterogeneity by organisation, specialty, and utilisation intensity of exchange functions. The qualitative strand will then examine how technical readiness (for example, data completeness and record timeliness), organisational readiness (for example, training and workflow fit), and trust and governance (for example, confidence in data provenance and consent practices) shape use and outcomes. The most important contribution to the mixed-methods synthesis will be a joint interpretation that maps the observed heterogeneity in effect sizes to these socio-technical mechanisms, thereby distinguishing ‘integration available’ from ‘integration used’.

Methodology overview

Methodologically, I propose a convergent mixed-methods design, integrating quantitative quasi- experimental evaluation with qualitative inquiry. The quantitative strand will use difference-in- differences and interrupted time series analyses over monthly panels across at least twenty-four months spanning the period of national integration announcements in January 2023. The analysis will compare outcomes in organisations with earlier versus later effective integration or with higher versus lower utilisation intensity, enabling estimation of integration-associated changes beyond background trends. (Department of Health – Abu Dhabi, 2023; WAM, 2023)

The quantitative strand will estimate the integration-associated changes in outcomes.

It will use difference-in-differences and interrupted time series analyses.

The qualitative strand will explain mechanisms, barriers, and enablers of effective use.

Data will be collected through semi-structured interviews and focus groups.

Findings will be integrated using a convergent mixed-methods design.

A joint interpretation will explain heterogeneity across organisations and time.

Priority for mixed-methods synthesis

When synthesising quantitative and qualitative evidence, the most important analytic component will be the explanation of heterogeneity in quantitative estimates. Specifically, I will prioritise (i) estimating effect sizes for continuity, safety, and efficiency outcomes alongside measures of HIE utilisation intensity, and (ii) using qualitative findings to explain why similar technical integration produces different patterns of use across settings. The synthesis will be presented through joint displays that align outcome changes, usage patterns, and thematic mechanisms. This approach will support credible interpretation of causal estimates and generate actionable recommendations for governance and implementation.

Quantitative strand: data and outcomes

The quantitative data will comprise de-identified extracts from participating organisations, including encounter data, diagnostic orders, medication orders, and HIE query or access logs. The primary outcome measures will be the rate of repeated laboratory tests and imaging within thirty days across different providers, emergency re-attendance within seventy-two hours, and thirty-day readmission. Where available, proxy safety indicators will include medication reconciliation discrepancies or duplicate prescribing flags. (Menachemi and Collum, 2011; Hong et al., 2018)

The analytic models will include organisation and time fixed effects, adjustment for seasonality and observable confounders, and sensitivity analysis using propensity-score weighting where feasible. (Menachemi and Collum, 2011; Hong et al., 2018)

Qualitative strand

The qualitative strand will use semi-structured interviews and focus groups with clinicians, informatics leads and governance stakeholders across Dubai and Abu Dhabi, as well as a small patient sample who have experienced cross-emirate care transitions. The purpose is to understand how HIE integration is used in practice, which information is trusted, how workflows change, and how privacy concerns are managed. Reflexive thematic analysis will be conducted, mapping emergent themes to established socio-technical constructs such as effort expectancy, facilitating conditions and trust, thereby providing explanatory leverage for quantitative heterogeneity. (Alkaabi and Elsori, 2025; Almarzouqi et al., 2022)

Computing artefact

A key computing contribution of this proposal is the creation of an artefact: an Interoperability and Data Quality Dashboard. This prototype will calculate and visualise indicators such as completeness of critical clinical data fields, timeliness of record availability, and concordance between local records and exchanged summaries. It is designed as a practical tool that can be used by digital health governance teams to identify improvement opportunities, monitor integration performance and communicate progress to stakeholders. The artefact will be developed iteratively using user- centred methods, and evaluated for technical validity and perceived usefulness through structured stakeholder feedback, drawing on IS acceptance constructs. (NABIDH, 2025; Department of Health – Abu Dhabi, 2019; Almarzouqi et al., 2022)

It will measure the completeness of critical clinical data fields.

It will assess the timeliness of record availability across organisations.

It will examine concordance between local records and exchanged summaries.

Ethics and data protection

Ethically, the proposal is grounded in data protection, confidentiality and minimising risk to participants. Quantitative datasets will be de-identified or pseudonymised, with any re- identification key retained by the data controller. Interview participants will provide informed consent, and recruitment strategies will be designed to avoid coercion, particularly for patient participants. Data will be stored securely with encryption and controlled access, and reporting will use aggregated findings with anonymised quotations. These measures align with the need to sustain trust in digital health platforms, a theme that recurs in UAE qualitative evidence. (Dubai Health Authority, 2021; Department of Health – Abu Dhabi, 2023; Alkaabi and Elsori, 2025)

Feasibility and planning

In terms of feasibility and planning, the work is structured across twenty-four weeks. Early weeks focus on governance arrangements and ethics. The middle period is devoted to data extraction, modelling, and parallel qualitative fieldwork. Artefact development overlaps with analysis so that early insights inform dashboard requirements. The final period integrates quantitative and qualitative findings and prepares the final outputs, including dissertation-style reporting and the narrated presentation artefacts. (Hong et al., 2018)

The full project is planned over twenty-four weeks.

In the early weeks, I will establish governance arrangements and secure ethics approvals.

In the middle period, I will extract data, run statistical models, and conduct qualitative fieldwork in parallel.

In the final period, I will integrate findings, finalise the dashboard artefact, and prepare the dissertation-style outputs and narrated materials.

Defining significant impact

To strengthen practical relevance, a ‘significant impact’ will be defined ex ante as sustained and statistically robust improvements that are also meaningful to stakeholders. Indicative thresholds are: at least a 10 per cent reduction in duplicated laboratory tests within thirty days across different providers; at least a 5 per cent reduction in duplicated imaging within thirty days; and a 3–5 per cent reduction in seventy-two-hour emergency re-attendance or thirty-day readmission, where these outcomes are available and appropriate for case-mix. For patient safety proxies, a meaningful impact would include a 10 per cent reduction in medication reconciliation discrepancies or duplicate prescribing flags, subject to data availability and validation. These thresholds will be refined with clinical and governance stakeholders during the feasibility phase.

Conclusion

To conclude, the UAE has made substantial progress in digital health infrastructure, but the discipline requires stronger evidence on whether interoperability at scale delivers measurable improvements in continuity, safety and efficiency. This proposal addresses that gap with a mixed-methods, quasi-experimental evaluation and a practical computing artefact designed to support governance. The anticipated contribution is both scholarly, through stronger causal inference and socio-technical explanation, and applied, through a reusable dashboard prototype that supports continuous improvement in HIE-enabled care. (Department of Health – Abu Dhabi, 2023; Abdulrahman et al., 2024; Almarzouqi et al., 2022)

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